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MODULE 2

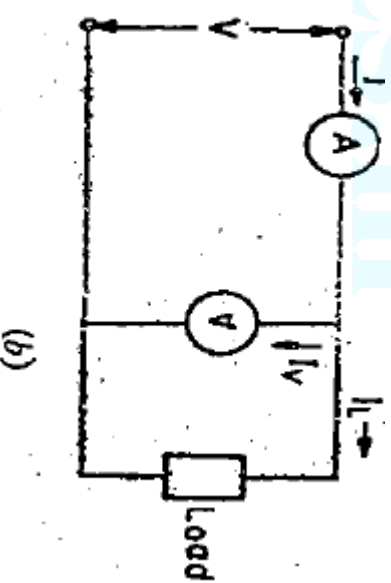
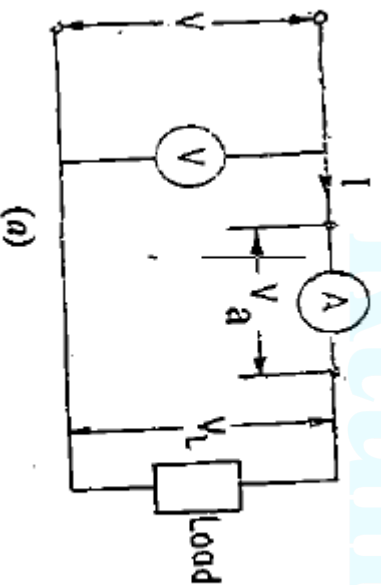
SYLLUBUS

- Measurement of power: Dynamometer type wattmeter – Construction and working - 3-phase power measurement-Low Power factor wattmeters.
- Measurement of energy: Induction type watt-hour meters-Single phase energy meter –construction and working, two element three phase energy meters ,
- Digital Energy meters -Time of Day(TOD) and Smart metering (description only).
- Current transformers and potential transformers – principle of working -ratio and phase angle errors.
- Extension of range using instrument transformers, Hall effect multipliers.

MEASUREMENT OF POWER

➤ Power in DC circuit

- The power taken by a load from a d.c. supply is given by the product of readings of an ammeter and a voltmeter when connected in the circuit as shown in the figure below.



- Power $P = V \times I$ watt

MEASUREMENT OF POWER

- The ammeter and voltmeter require power for their operation and allowance must be made for this.
- Either of the two connections shown in Figs (a) and (b) may be used.
- In Fig (a), the ammeter is connected between load and voltmeter. Therefore, the voltmeter not only indicates the voltage V_L across the load, but in addition voltage drop $V_a = I_a R_a$ across the ammeter.

- Power consumed by load = $V_L I = (V - V_a) I$

$$= VI - V_a I = VI - I^2 R_a$$

= power indicated by instruments -
power loss in ammeter.

MEASUREMENT OF POWER

- In Fig b, voltmeter is connected between load and ammeter. Therefore, ammeter not only indicates the current through the load but in addition current in the voltmeter also.
- Current through the voltmeter $I_v = V/R_v$ where R_v = resistance of voltmeter.
- Power consumed by load $= V I_L = V(I - I_v) = V(I - (V/R_v))$

$$= VI - V^2/R_v$$

= power indicated by instruments -
power loss in Voltmeter

MEASUREMENT OF POWER

- Thus in both the cases, the power indicated by the instruments is equal to the power consumed by the load plus the power consumed by the instrument nearest to the load terminals.
- In order to obtain the true power, corrections must be applied for power loss in instruments.
- Under normal conditions the value of power loss in instruments is quite small compared with the load power and therefore the error introduced is small.

MEASUREMENT OF POWER

➤ Power in A.C. circuits

- In the case of alternating currents, the instantaneous power varies continuously as the current and voltage go through a cycle.
- However, we are not interested in the instantaneous power but in its **average value over a cycle**.

MEASUREMENT OF POWER

The instantaneous power is given by $p=vi$

Thus, if both current and voltage waves are sinusoidal, the current lagging in phase by an angle ϕ , then

$$v=V_m \sin \omega t \quad \text{and } i=i_m \sin(\omega t - \phi)$$

Therefore instantaneous power is given by $p=vi=V_m \sin \omega t \, i_m \sin(\omega t - \phi)$

if $\theta = \omega t$,

$$p = V_m \sin \theta \, i_m \sin(\theta - \phi)$$

$$\begin{aligned} \text{Average power over a cycle is } P &= \frac{V_m i_m}{2\pi} \int_0^{2\pi} \cos \phi - \cos(2\theta - \phi) d\theta = \frac{V_m i_m}{2} \cos \phi \\ &= V I \cos \phi \end{aligned}$$

Where V and I are rms values of voltage and current. $\cos \phi$ is the power factor of the load

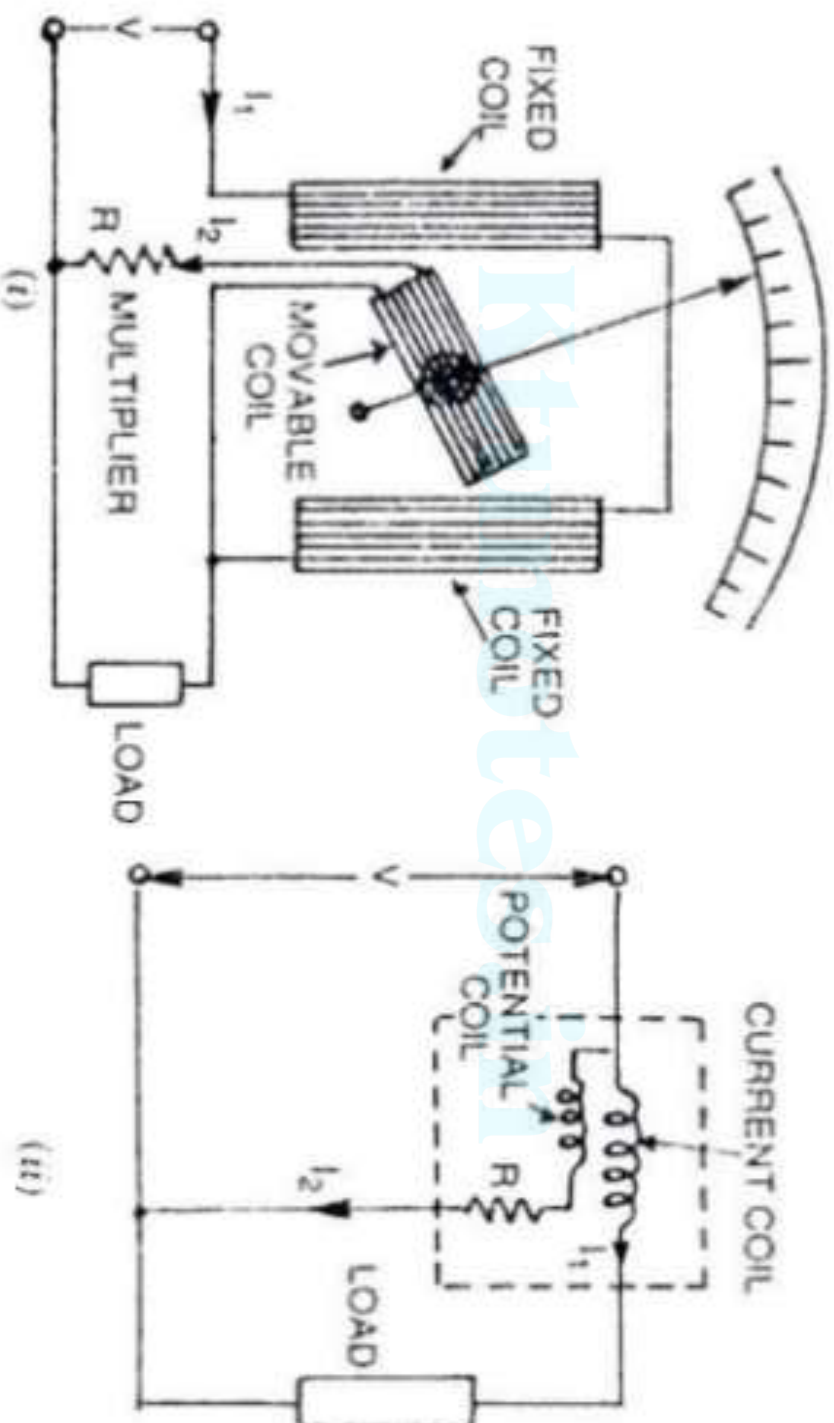
MEASUREMENT OF POWER

- Since the power factor ($\cos\phi$) is involved in the expression for power means that a **wattmeter** must be used for measurement of power in a.c. circuits instead of merely an ammeter and a voltmeter, since the later method takes no account of the power factor.

ELCTRODYNAMOMETER WATTMETERS

- This instrument consists of two coils namely fixed and moving coil connected in different circuits and its working depends on the reaction between the magnetic field produced by moving and fixed coils.
- The fixed coils or "field coils" are connected in series with the load and carry the current in the circuit. The fixed coils, therefore, form the "current coil" of-the wattmeter.
- The moving coil is connected across the voltage and therefore, carries a current proportional to the voltage. Since the moving coil carries a current proportional to the voltage, It is called the "pressure coil" or "voltage coil" of the wattmeter.

ELECTRODYNAMOMETER WATTMETERS



ELECTRODYNAMOMETER WATTMETERS

➤ Construction

➤ Fixed Coils

- The fixed coils carry the current of the circuit.
- They are divided into two halves.
- The fixed coils are wound with heavy wire.
- This wire is stranded or laminated especially when carrying heavy current in order to avoid eddy current losses in conductors.

ELCTRODYNAMOMETER WATTMETERS

➤ Moving Coil

- It is mounted on a pivoted spindle and is entirely embraced by the fixed current coils.
- Since the current of the moving coil is carried by the instrument springs, it is limited to values, which can be carried safely by springs.

➤ Control.

- Spring control is used for the instrument

ELECTRODYNAMOMETER WATTMETERS

➤ Damping

- Air friction damping is used.
- The moving system carries a light aluminium vane which moves in a sector shaped box.

➤ Scales and Pointers

- They are equipped with mirror type scales and knife edge pointers to remove reading errors due to parallax.

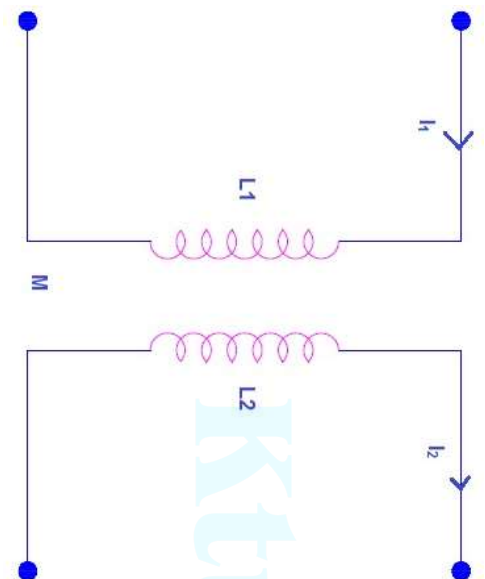
ELTCTRODYNAMOMETER WATTMETERS

➤Working principle

When the wattmeter is connected in the circuit to measure the power, the current coil carries load current and potential coil carries current proportional to the load voltage, due to currents in the coils, mechanical force exists between them. The result is that movable coil moves the pointer over the scale. The pointer comes to rest at a position where deflecting torque is equal to the controlling torque (reversals of current reverses current's in both the fixed coils and the movable coil so that the direction of deflecting torque remains unchanged).

ELECTRODYNAMOMETER WATTMETERS

➤ Torque Equation in Electrodynamometer Instruments



- Let us assume that the current in fixed coil be i_1 and that in moving coil be i_2 as shown in figure below.
- Also assume that,

L_1 = Self inductance of fixed coil

L_2 = Self inductance of moving coil

M = Mutual inductance between fixed and moving coils

• Thus,

The flux linkage of fixed coil $\Phi_1 = L_1 i_1 + M i_2$

The flux linkage of moving coil $\Phi_2 = L_2 i_2 + M i_1$

The electrical energy input to the instrument,

$$= e_1 i_1 dt + e_2 i_2 dt$$

ELTCTRODYNAMOMETER WATTMETERS

- But according to [Faraday's Law](#),

$$e_1 = d\Phi_1/dt$$

$$\text{and } e_2 = d\Phi_2/dt$$

- Therefore energy input to the instrument

$$\begin{aligned} &= i_1 d\Phi_1 + i_2 d\Phi_2 \\ &= i_1 d(L_1 i_1 + M i_2) + i_2 d(L_2 i_2 + M i_1) \\ &= i_1 L_1 di_1 + i_1^2 dL_1 + i_1 i_2 dM + i_1 M di_2 + i_2 L_2 di_2 + \\ &\quad i_2^2 dL_2 + i_1 i_2 dM + i_2 M di_1 \end{aligned}$$

- Since L_1 and L_2 are constant, therefore $dL_1 = 0$ and $dL_2 = 0$
 $= i_1 L_1 di_1 + i_1 i_2 dM + i_1 M di_2 + i_2 L_2 di_2 + i_1 i_2 dM + i_2 M di_1 \dots (1)$

ELCTRODYNAMOMETER WATTMETERS

- Some of the above input energy to electrodynanometer instruments are stored in the form of magnetic energy in the coil while rest is converted into mechanical energy of moving coil.
- Thus we can write,
$$\text{Energy Input} = \text{Mechanical Energy} + \text{Stored Energy}$$
$$\text{Mechanical Energy} = \text{Electrical Input} - \text{Stored Energy} \dots(2)$$
- Thus to find the mechanical energy, we need to find the change in stored energy in the magnetic field of the coil. Let us assume an infinitesimally small time dt for the sake of calculation of change in stored energy.

ELTCTRODYNAMOMETER WATTMETERS

Change in stored energy = $d(1/2 L_1 i_1^2 + 1/2 L_2 i_2^2 + M i_1 i_2)$

$$= i_1 L_1 di_1 + i_2 L_2 di_2 + i_1 M di_2 + i_2 M di_1 + i_1 i_2 dM + (i_1^2/2) dL_1 + (i_2^2/2) dL_2$$

- But L_1 and L_2 are constant, therefore $dL_1 = 0$ and $dL_2 = 0$

$$\text{Change in stored energy} = i_1 L_1 di_1 + i_2 L_2 di_2 + i_1 M di_2 + i_2 M di_1 + i_1 i_2 dM \dots\dots (3)$$

- From equation (1), (2) and (3),

$$\text{Mechanical Energy} = i_1 i_2 dM$$

ELTCTRODYNAMOMETER WATTMETERS

- Let T_d be the deflecting torque and $d\theta$ be the change in deflection, then

mechanical energy = $T_d d\theta$

$$T_d d\theta = i_1 i_2 dM$$

$$T_d = i_1 i_2 \frac{dM}{d\theta}$$

- The above equation gives the deflecting torque in electro-dynamics or electro-dynamometer instruments.
- It can be seen that deflecting torque depends upon the multiplication of instantaneous value of current and rate of change of mutual inductance between the fixed and moving coil.

ELECTRODYNAMOMETER WATTMETERS

11.3.2. Theory of Electrodynamometer Wattmeters. The instantaneous torque of an electrodynamic instrument is given by :

$$T_i = i_1 i_2 \frac{dM}{d\theta}$$

where i_1 and i_2 are instantaneous values of currents in two coils. Let V and I be the r.m.s. values of voltage and current being measured.

$\therefore$ Instantaneous value of voltage across the pressure coil circuit

$$v = \sqrt{2} V \sin \omega t.$$

If the pressure coil circuit has a very high resistance, it can be treated as purely resistive. Therefore, current i_p in the pressure coil is in phase with the voltage and its instantaneous value is :

$$i_p = v/R_p = \sqrt{2} V/(R_p) \sin \omega t = \sqrt{2} I_p \sin \omega t$$

$$I_p = V_p/R_p = \text{rms value of current in pressure coil circuit.}$$

here
and

R_p = resistance of pressure coil circuit.

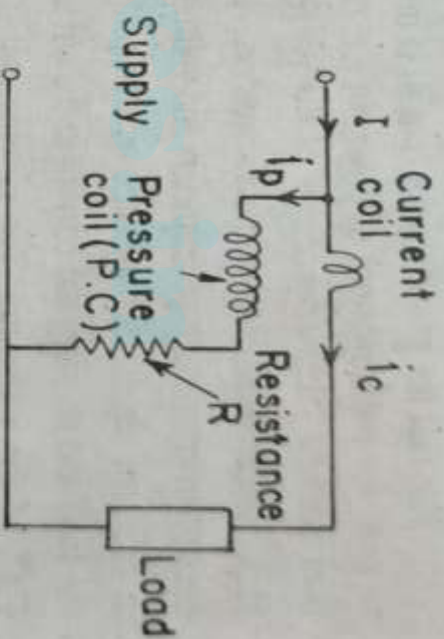


Fig. 11.3. Circuit of electrodynamic wattmeter.

ELECTRODYNAMOMETER WATTMETERS

If the current in the current coil lags the voltage in phase by an angle ϕ , instantaneous value of current through current coil is :

$$i_c = \sqrt{2} I \sin (\omega t - \phi)$$

$\therefore$ Instantaneous torque

$$T_i = \sqrt{2} I_p \sin \omega t \times \sqrt{2} I \sin (\omega t - \phi) \frac{dM}{d\theta}$$

$$= 2 I_p I \sin \omega t \sin (\omega t - \phi) \frac{dM}{d\theta} \approx I_p I (\cos \phi - \cos (2\omega t - \phi)) \frac{dM}{d\theta}$$

...(11.4)

It is clear from above that there is a component of power which varies as twice the frequency of current and voltage (mark the term containing $2\omega t$).

Average deflecting torque

$$T_d = \frac{1}{T} \int_0^T T_i d(\omega t) = \frac{1}{T} \int_0^T I_p I [\cos \phi - \cos (2\omega t - \phi)] \frac{dM}{d\theta} \cdot d(\omega t)$$

$$= I_p I \cos \phi \cdot \frac{dM}{d\theta}$$

...(11.5)

$$= (V/I R_p) \cos \phi \cdot \frac{dM}{d\theta}$$

...(11.6)

Controlling torque exerted by springs $T_c = K\theta$

where

K = spring constant and θ = final steady deflection.

ELECTRODYNAMOMETER WATTMETERS

Since the moving system of the instrument cannot follow the rapid variations in torque (the torque has a double frequency component), it will take up a position at which the average deflection torque is equal to the restoring torque of the springs.

∴ At balance position or deflection

$$K\theta = I_p I \cos \phi \cdot dM/d\theta$$

$$\theta = I_p I \cos \phi (dM/d\theta)/K \quad \dots(11.7)$$

$$= (VI \cos \phi / R_p K) dM/d\theta \quad \dots(11.8)$$

$$= K_1 VI \cos \phi \cdot dM/d\theta \quad \dots(11.9)$$

$$= (K_1 dM/d\theta) P \quad \dots(11.10)$$

$$P = \text{power being measured} = VI \cos \phi$$

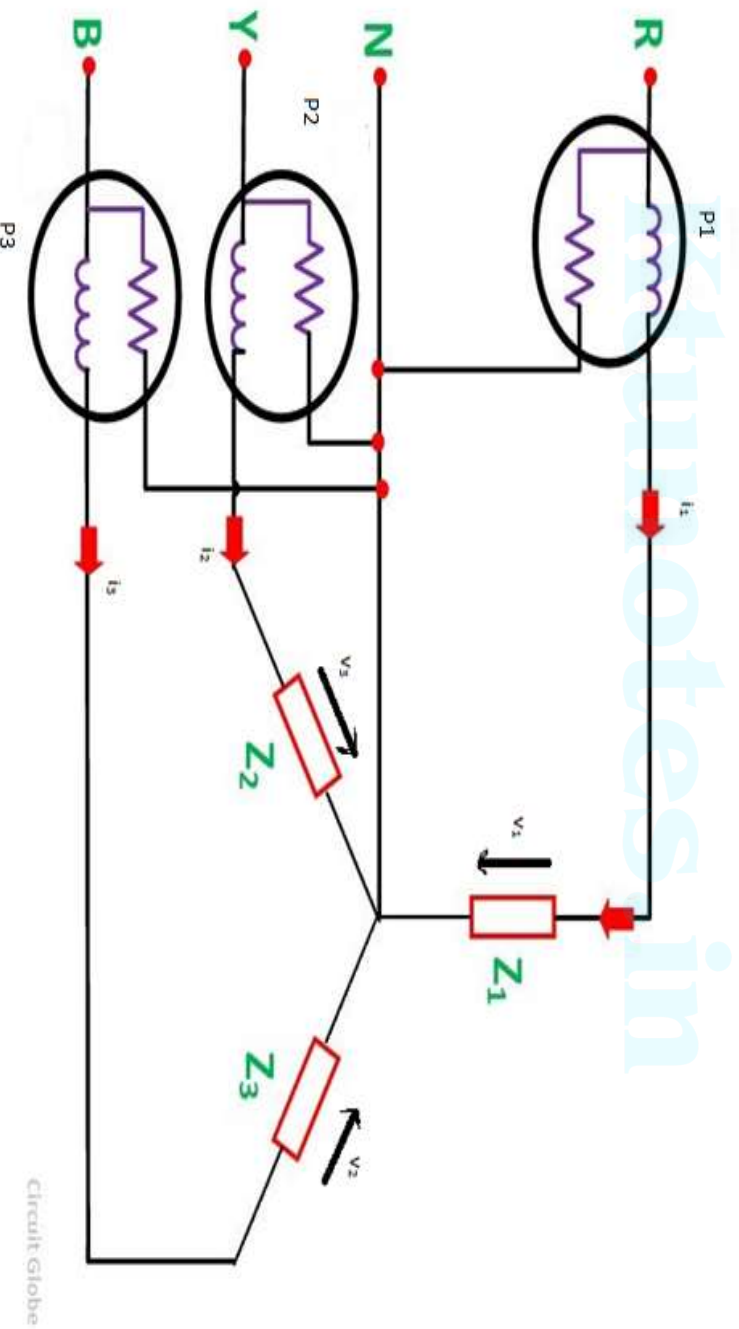
$$K_1 = 1/R_p K \quad \dots(11.11)$$

where
and

MEASUREMENT OF POWER IN THREE PHASE CIRCUITS

➤ Three wattmeter method

- Three wattmeter method employed to measure power in a 3-phase, 4 wire system is shown below.



MEASUREMENT OF POWER IN THREE PHASE CIRCUITS

- Current element of a wattmeter is connected in each line and the corresponding voltage element is connected between the line and neutral.
- The total power is measured by summing the readings of 3 wattmeters .
- Sum of the instantaneous readings of wattmeters ,

$$P = P_1 + P_2 + P_3 = V_1 I_1 + V_2 I_2 + V_3 I_3$$

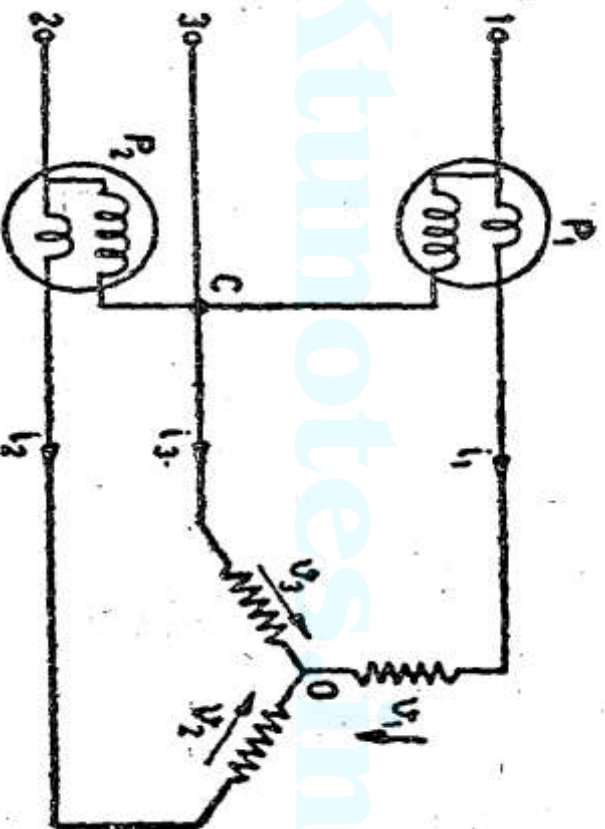
MEASUREMENT OF POWER IN THREE PHASE CIRCUITS

➤ Two wattmeter method

- In a three phase three wire system we require 3 elements. But if we make the common points of the pressure coils coincide with one of the lines, then we will require only $n-1 = 2$ elements .

MEASUREMENT OF POWER IN THREE PHASE CIRCUITS

➤ Star connected system



Instantaneous power consumed by load = $v_1 i_1 + v_2 i_2 + v_3 i_3$

Instantaneous reading of P_1 wattmeter $p_1 = i_1 (v_1 - v_3)$

Instantaneous reading of P_2 wattmeter $p_2 = i_2 (v_2 - v_3)$

MEASUREMENT OF POWER IN THREE PHASE CIRCUITS

Sum of instantaneous readings of two wattmeters = $p_1 + p_2$

$$= i_1 (v_1 - v_3) + i_2 (v_2 - v_3) \\ = v_1 i_1 + v_2 i_2 - v_3 (i_1 + i_2)$$

From Kirchhoff's law,

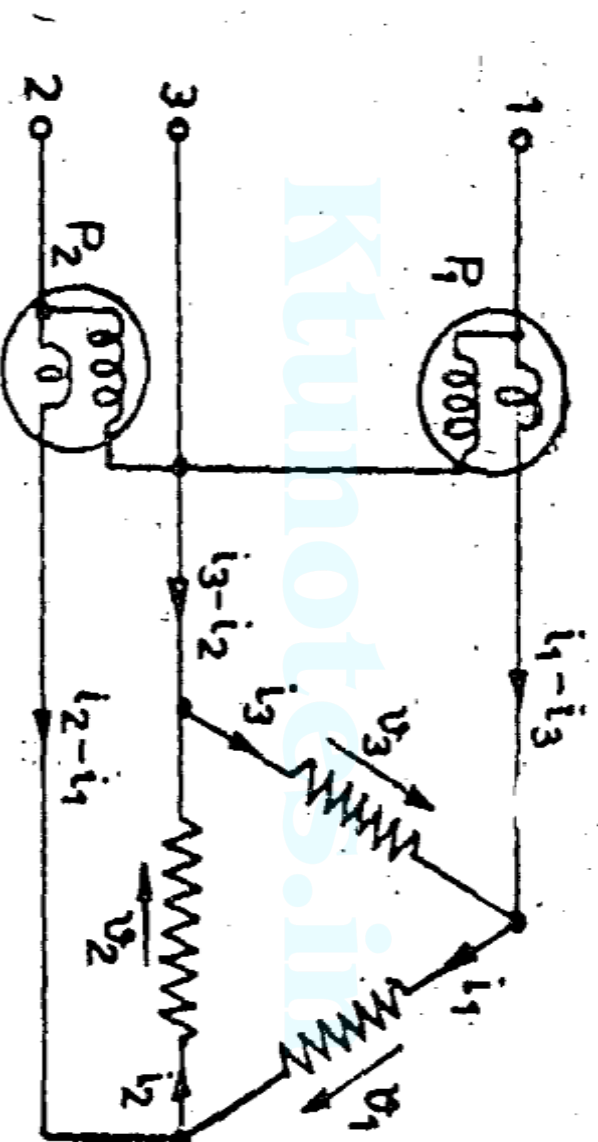
$$i_1 + i_2 + i_3 = 0 \quad \text{or} \quad i_3 = -(i_1 + i_2)$$

Sum of instantaneous readings of two wattmeters = $v_1 i_1 + v_2 i_2 + v_3 i_3$

- Therefore, the sum of the two wattmeter reading is equal to the power consumed by the load.
- This is irrespective of whether the load is balanced or unbalanced.

MEASUREMENT OF POWER IN THREE PHASE CIRCUITS

➤ Delta connection



Instantaneous power consumed by load = $v_1 i_1 + v_2 i_2 + v_3 i_3$

Instantaneous reading of p_1 wattmeter $p_1 = -v_3 (i_1 - i_3)$

Instantaneous reading of p_2 wattmeter $p_2 = v_2 (i_2 - i_1)$

MEASUREMENT OF POWER IN THREE PHASE CIRCUITS

- Sum of instantaneous readings of wattmeters p_1 and p_2

$$p_1 + p_2 = -v_3 (i_1 - i_3) + v_2 (i_2 - i_1)$$

$$= v_2 i_2 + v_3 i_3 - i_1 (v_2 + v_3)$$

- From Kirchhoff's voltage law,

$$v_1 + v_2 + v_3 = 0 \quad \text{or} \quad v_1 = -(v_2 + v_3)$$

Hence sum of instantaneous readings of wattmeters

$$= v_2 i_2 + v_3 i_3 - i_1 (-v_1) = v_1 i_1 + v_2 i_2 + v_3 i_3$$

- Therefore, the sum of the two wattmeter reading is equal to the power consumed by the load.
- This is irrespective of whether the load is balanced or unbalanced.

MEASUREMENT OF POWER IN THREE PHASE CIRCUITS

- Figure shows the phasor diagram for a balanced star connected load.
- Let V_1 , V_2 , V_3 be the rms values of phase voltage and I_1 , I_2 , I_3 be the rms values of phase currents.
- The load is balanced, therefore,

$$\text{Phase voltages } V_1 = V_2 = V_3 = V$$

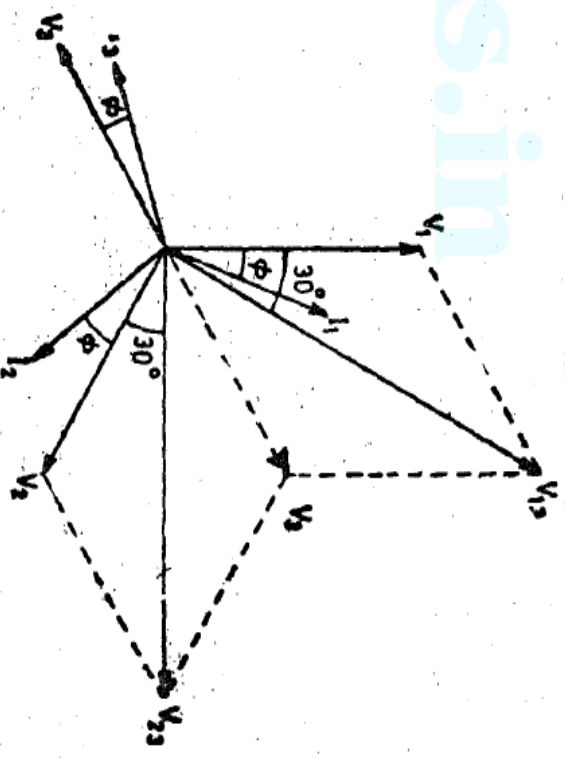
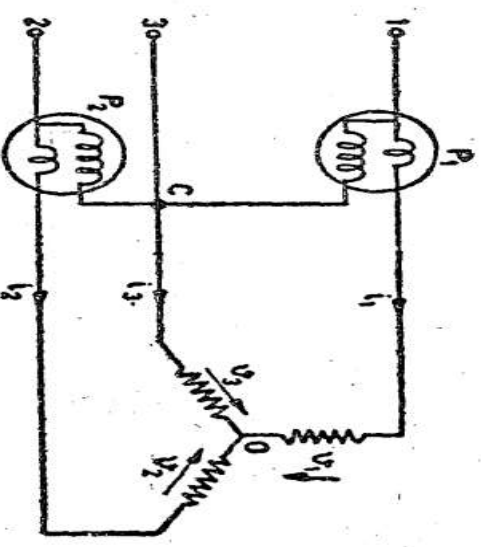
$$\text{Line voltages } V_{13} = V_{23} = V_{12} = \sqrt{3} V$$

$$\text{Phase currents } I_1 = I_2 = I_3 = I$$

$$\text{Line currents } I_1 = I_2 = I_3 = I$$

$$\text{Power factor} = \cos \phi$$

- The phase currents lag the corresponding phase voltage by an angle ϕ



MEASUREMENT OF POWER IN THREE PHASE CIRCUITS

- The current through wattmeter P1 is I_1 and voltage across its pressure coil is V_{13} . I_1 leads V_{13} by an angle $(30-\phi)$.

Reading of P1 wattmeter, $P1 = V_{13} I_1 \cos(30-\phi)$

$$= \sqrt{3} VI \cos(30-\phi)$$

- The current through wattmeter P2 is I_2 and voltage across its pressure coil is V_{23} . I_2 lags V_{23} by an angle $(30+\phi)$.

Reading of P2 wattmeter, $P2 = V_{23} I_2 \cos(30+\phi)$

$$= \sqrt{3} VI \cos(30+\phi)$$

Sum of reading of two wattmeters,

$$P1 + P2 = \sqrt{3} VI [\cos(30-\phi) + \cos(30+\phi)] = 3VI \cos \phi$$

- This is the total power consumed by the load.

MEASUREMENT OF POWER IN THREE PHASE CIRCUITS

Now $P_1 - P_2 = \sqrt{3} V_I [\cos(30^\circ - \phi) - \cos(30^\circ + \phi)] = \sqrt{3} V_I \sin \phi$

$$\frac{P_1 - P_2}{P_1 + P_2} = \frac{\sqrt{3} V_I \sin \phi}{3 V_I \cos \phi} = \frac{\tan \phi}{\sqrt{3}}$$

$$\phi = \tan^{-1} \sqrt{3} \frac{P_1 - P_2}{P_1 + P_2}$$

$$\text{power factor, } \cos \phi = \cos \left(\tan^{-1} \sqrt{3} \frac{P_1 - P_2}{P_1 + P_2} \right)$$

MEASUREMENT OF POWER IN THREE PHASE CIRCUITS

➤ Effect of power factor on the readings of wattmeters

- Case 1- with unity p.f. $\cos\phi=1$ and $\phi=0$

The readings of the two wattmeters are:

$$P_1 = \sqrt{3} V I \cos(30-\phi) = \sqrt{3} V I \cos 30 = (3/2) V I$$

$$P_2 = \sqrt{3} V I \cos(30+\phi) = \sqrt{3} V I \cos 30 = (3/2) V I$$

$$P_1 + P_2 = 3 V I$$

- The readings of the two wattmeters are equal and each wattmeter reads half of the power

- Case 2-When $p.f=0.5$, $\phi=60^\circ$
- Case 3-when $p.f=0$, $\phi=90^\circ$

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MEASUREMENT OF ENERGY

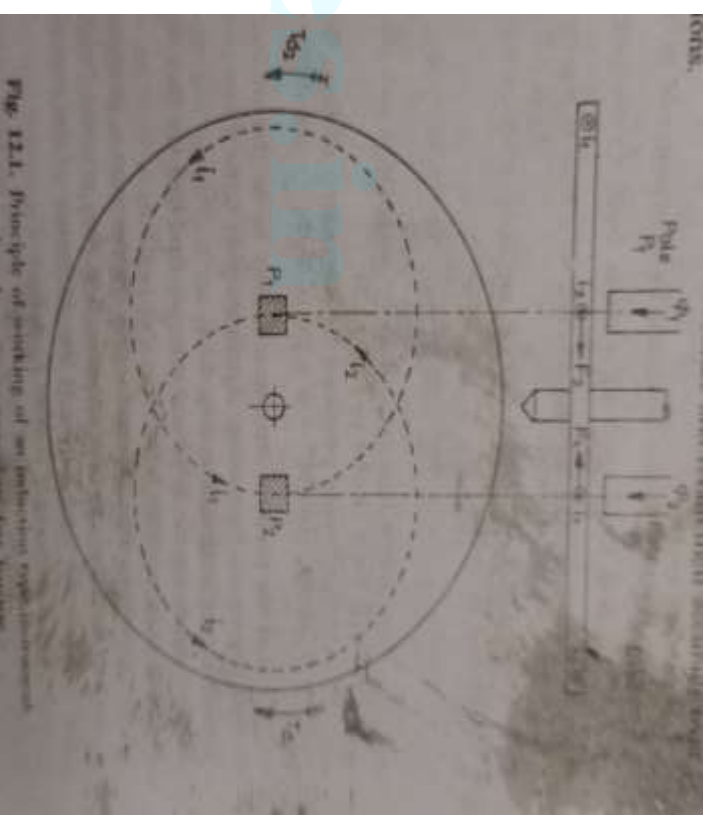
- Energy is the total power delivered or consumed over a time interval.
- i.e. energy=power x time
- Electrical energy developed as work or dissipated as heat over an interval of time t is,

$$W = \int_0^t P \, dt$$

- Unit of energy is Joule or watt-second
- Since watt-second is of inconvenient size, a larger unit-kilowatt hour is used

THEORY OF INDUCTION TYPE METERS

- In all induction type instruments there are two fluxes produced by the current flowing in the windings of the instrument
- These are alternating fluxes, so they produce an emf in a metallic disc provided in the instrument.
- These emfs circulate eddy currents in the metallic disk.
- Since there are two fluxes and two eddy currents, to torques will be produced.
 - One due to the interaction of first flux and eddy current produced by the second flux.
 - Second due to the interaction of second flux and eddy current produced by the first flux.



THEORY OF INDUCTION TYPE METERS

Let ϕ_1 and ϕ_2 be the instantaneous values of two fluxes having a phase difference of β and, therefore, we can write :

$$\phi_1 = \Phi_{m1} \sin \omega t, \quad \phi_2 = \Phi_{m2} \sin (\omega t - \beta)$$

$$\Phi_1 = \Phi_{m1} / \sqrt{2} \quad \text{and} \quad \Phi_2 = \Phi_{m2} / \sqrt{2}$$

Their rms values are

Flux Φ_1 produces an emf in the disc by transformer action.

The instantaneous value of this emf is,

$$e_1 \propto \left[-\frac{d(\Phi_1)}{dt} \right] \propto \left[-\frac{d}{dt} (\Phi_{m1} \sin \omega t) \right], \quad \propto [-\omega \Phi_{m1} \cos \omega t], \quad \propto [-f \Phi_{m1} \cos \omega t]$$

It is clear from above that e_1 lags the flux by 90°

Let $E_1 = \text{rms value of emf } e_1,$

$$\therefore E_1 \propto f \frac{\Phi_{m1}}{\sqrt{2}} \propto f \Phi_1 \quad \dots (12.7)$$

This emf circulates an eddy current having an rms value

I_1 . If the impedance of eddy current path is Z ,

$$I_1 = \frac{E_1}{Z} \propto \frac{f}{Z} \Phi_1.$$

I_1 lags the voltage E_1 by an angle α where $\tan \alpha$ is the ratio of reactance to resistance of path of eddy currents in the disc.

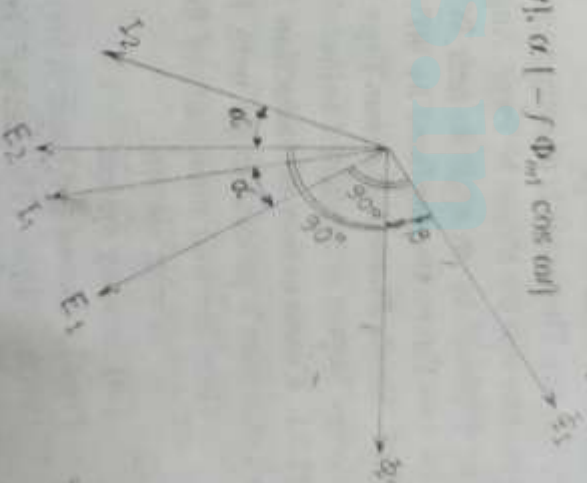


Fig. 12.2 Phasor diagram of an induction type instrument.

THEORY OF INDUCTION TYPE METERS

Referring to phasor diagram shown in Fig. 12.2 the phase angle between Φ_2' and I_1 is $(90^\circ - \beta + \alpha)$.

$\therefore$ Average torque produced by interaction of Φ_2 and I_1 is,

$$T_{d1} \propto \Phi_2 I_1 \cos (90^\circ - \beta + \alpha), \propto \Phi_2 \Phi_1 \frac{f}{Z} \cos (90^\circ - \beta + \alpha). \quad \dots(12.8)$$

Now take flux Φ_2 and proceed in a similar manner. It produces an eddy emf of rms value is,

$$E_2 \propto f \Phi_2.$$

E_2 lags the flux Φ_2 by 90° , E_2 circulates eddy currents in the same disc. The rms value of eddy current produced by emf E_2 is :

$$I_2 \propto \frac{f}{Z} \Phi_2.$$

This current lags the emf E_2 by an angle α and, therefore, the angle between flux Φ_1 and current I_2 is $(90^\circ + \beta + \alpha)$. (Refer to Fig. 12.2).

Average torque produced by interaction of flux, Φ_1 and current I_2 is :

$$T_{d2} \propto \Phi_1 I_2 \cos (90^\circ + \alpha + \beta), \propto \Phi_1 \Phi_2 \frac{f}{Z} \cos (90^\circ + \alpha + \beta). \quad \dots(12.9)$$

THEORY OF INDUCTION TYPE METERS

Let the positive directions of flux be as shown and assume that, at the instant under consideration, both fluxes are increasing. Application of Lenz's law shows that the directions of currents i_1 and i_2 will be those as indicated in Fig. 12.1.

Applying the left hand rule, we find that the flux Φ_2 interacting with current i_1 produces a force F_1 acting from right to left.

Similarly flux Φ_1 interacting with current i_2 produces a force F_2 acting from left to right and since the two forces are in opposite direction, they produce two torques T_{d1} and T_{d2} acting in different directions.

Thus the total torque is the difference of the two torques or

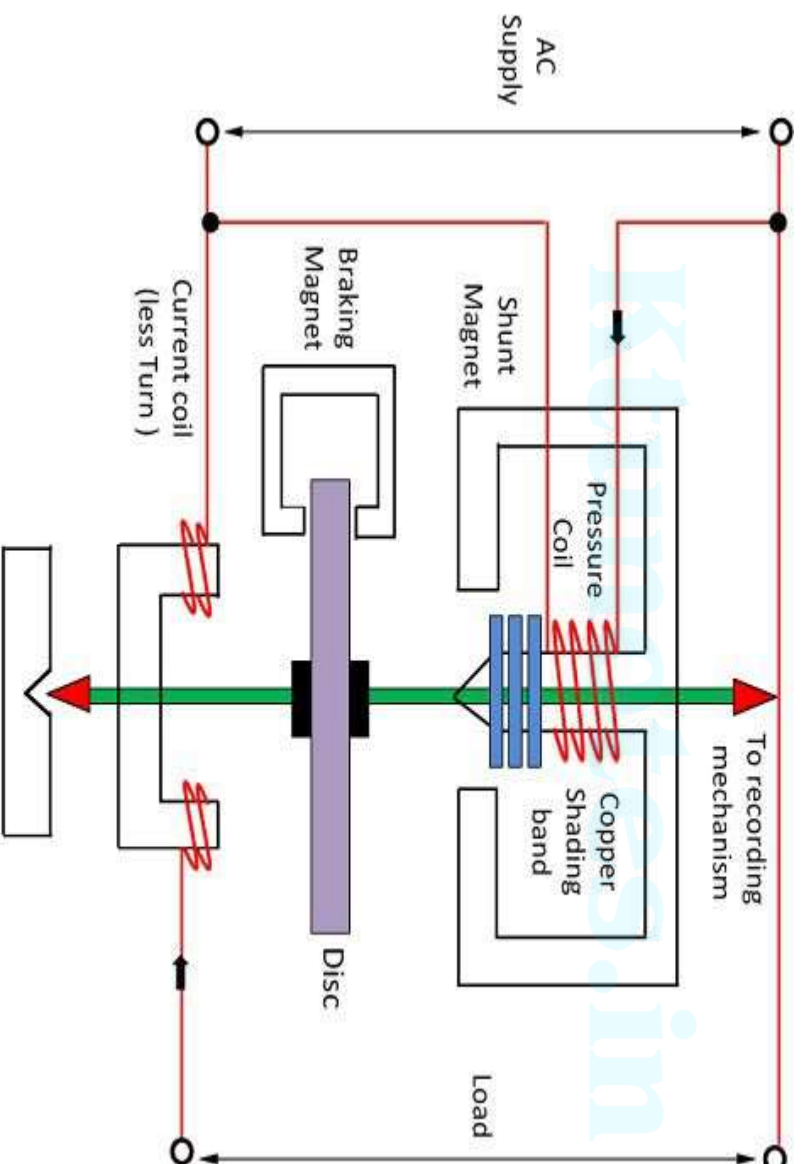
Total deflecting torque $T_d = T_{d1} - T_{d2}$

$$\propto \Phi_2 \Phi_1 \frac{f}{Z} \cos (90^\circ - \beta + \alpha) - \Phi_2 \Phi_1 \frac{f}{Z} \cos (90^\circ + \beta + \alpha)$$

$$\propto \Phi_2 \Phi_1 \frac{f}{Z} [\sin (\beta - \alpha) + \sin (\beta + \alpha)], \quad \alpha \Phi_1 \Phi_2 \frac{f}{Z} \sin \beta \cos \alpha \quad \dots (12.10)$$

SINGLE PHASE INDUCTION TYPE ENERGY METERS

- Induction type of energy meters are universally used for measurement of energy in domestic and industrial **ac** circuits.



Induction Type Energy Meter

SINGLE PHASE INDUCTION TYPE ENERGY METERS

➤ Construction

- Four main parts

➤ Driving System:-

- It consist of two electromagnets
- The core of these electromagnets are made up of silicon steel laminations.
- The coil of one of the electromagnets is excited by the load current and the coil is known as current coil and magnet as series magnet.
- The coil of the other electromagnet is connected across the supply voltage and is thus excited by a current proportional to voltage.
- This coil is called as pressure coil and electromagnet as shunt magnet.
- Copper shading bands are provided on the central limb .The position of these bands are adjustable and its function is to bring shunt magnet flux exactly in quadrature with the applied voltage.

SINGLE PHASE INDUCTION TYPE ENERGY METERS

➤ Moving system:-

- This consist of an aluminium disc mounted on a light alloy shaft. It is positioned in the air gap between shunt and series magnet and is free to rotate.

➤ Braking system:-

- A permanent magnet positioned near the edge of aluminium disc forms the braking system.
- The aluminium disc moves in the field of this permanent magnet and thus provides braking torque.
- The position of the permanent magnet can be adjusted and therefore braking torque can be adjusted

SINGLE PHASE INDUCTION TYPE ENERGY METERS

- **Registering** (counting)mechanism
- Its function is to record continuously a number which is proportional to the revolutions made by the moving system.

SINGLE PHASE INDUCTION TYPE ENERGY METERS

➤ Theory of single phase energy meters

Let V =applied voltage,

I = load current,

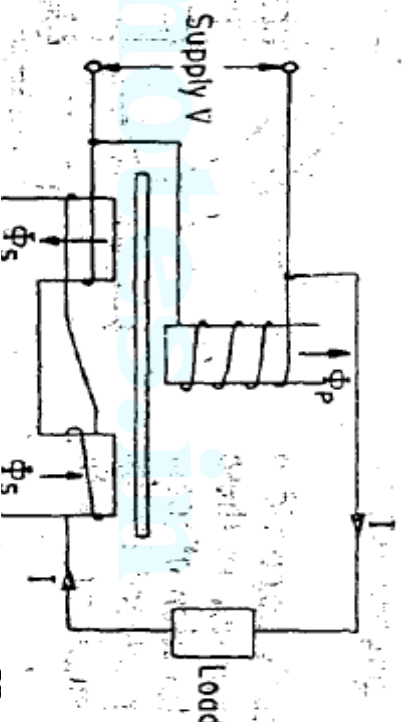
ϕ = phase angle of load,

I_p = pressure coil current,

Δ =phase angle between supply voltage and pressure coil flux. f =frequency,

Z = impedance of eddy current paths,

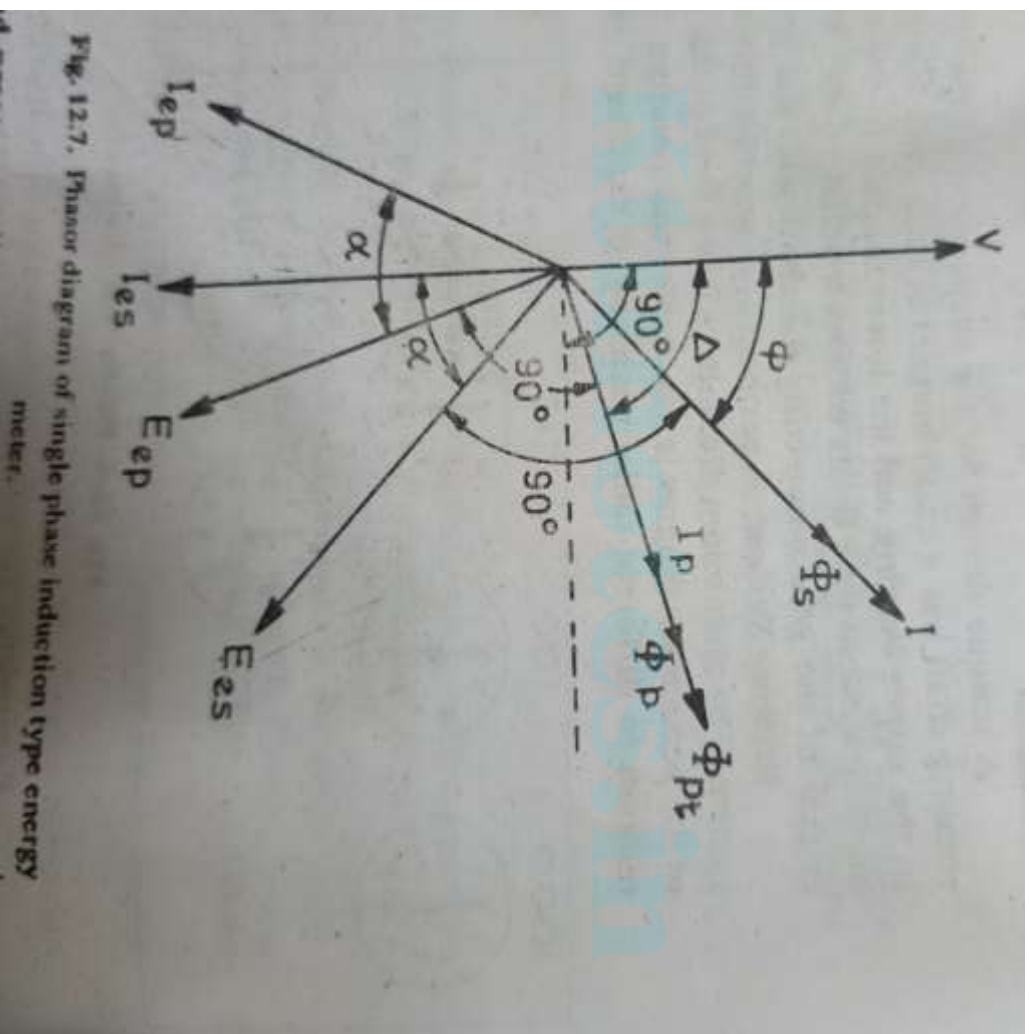
E_{ep} = eddy emf induced by flux Φ_p ,

 E_{es} = eddy emf induced by flux Φ_s , α = phase angle of eddy current paths,

I_{ep} = eddy current due to flux Φ_p ,

I_{se} = eddy current due to flux Φ_s .

SINGLE PHASE INDUCTION TYPE ENERGY METERS



SINGLE PHASE INDUCTION TYPE ENERGY METERS

$$T_d \propto \Phi_1 \Phi_2 \frac{f}{Z} \sin \beta \cos \alpha = K_1 \Phi_1 \Phi_2 \frac{f}{Z} \sin \beta \cos \alpha$$

where

$K_1 = \text{a constant.}$

Now $\beta = \text{phase angle between fluxes } \Phi_1 \text{ and } \Phi_2.$

In our case, the two fluxes are Φ_p and Φ_s ,

$\therefore \beta = \text{phase angle between fluxes } \Phi_p \text{ and } \Phi_s = (\Delta - \phi) \text{ (See Fig. 11.5)}$

$\therefore \text{Driving torque } T_d = K_1 \Phi_p \Phi_s \frac{f}{Z} \sin (\Delta - \phi) \cos \alpha$

But $\Phi_p \propto V$ and $\Phi_s \propto I.$

$\therefore T_d = K_2 V I \frac{f}{Z} \sin (\Delta - \phi) \cos \alpha$

SINGLE PHASE INDUCTION TYPE ENERGY METERS

If f , Z and α are constant, $T_d = K_3 VI \sin (\Delta - \phi)$

... (11.7)

If N is the steady speed, braking torque $T_B = K_4 N$.

At steady speed the driving torque must equal the braking torque

$$\therefore K_4 N = K_3 VI \sin (\Delta - \phi), N = K VI \sin (\Delta - \phi)$$

... (11.8)

$$\text{If } \Delta = 90^\circ, N = K VI \sin (90 - \phi) = K VI \cos \phi$$

... (11.9)

$$= K \times (\text{power}).$$

Thus, in order that the speed of rotation is proportional to power, angle Δ should be equal to 90° . Hence the flux Φ_s must be made to lag the supply voltage by exactly 90° .

$$\text{Total number of revolutions} = \int N dt = K \int VI \sin (\Delta - \phi) dt$$

... (11.10)

$$\text{If } \Delta = 90^\circ, \text{ total number of revolutions} = K \int (VI \cos \phi) dt$$

... (11.11)

$$= K \int (\text{power}) dt = K \times (\text{energy})$$

LAG ADJUSTMENTS IN ENERGY METER

- The meter will register true energy only if the angle between supply voltage and flux produced by shunt magnet is made equal to 90° .
- This requires that the pressure coil winding should be so designed that it is highly inductive and has a low resistance and the iron losses in the core are small.
- But even with this the phase of flux ϕ_p is not 90° with respect, to voltage V but a few degrees less than 90° .
- By introducing a magnetic shunt circuit which allows the main portion of the shunt magnet flux to bypass the gap in which the disc is located it is possible to introduce an mmf in the proper relationship to make the shunt magnet flux in exact quadrature with supply voltage.

LIGHT LOAD / FRICTION

COMPENSATION IN ENERGY METER

- In order to compensate frictional forces we have to apply small force in the direction of rotation of disc.
- This applied force should be independent of load, so that the meter can read correctly at light load also.
- But over compensation of friction leads to creeping.
- Creeping may be defined as the slow but continuous rotation of disc when there is no current flowing through the current coil and only the pressure coil is energized .
- In order to avoid creeping two holes are drilled, which are diametrically opposite to each other on the disc.
- Due to this, the effective circular eddy current path of the disc is distorted.

ERRORS IN SINGLE PHASE ENERGY METER

- Errors caused by driving system
 - **Incorrect magnitude of fluxes** :-This may be due to abnormal values of current or voltage. The shunt magnet flux may be in error due to changes in resistance of coil or due to abnormal frequencies.
 - **Incorrect phase angles**:- There may not be proper relationship between the various phasors . This may be due to improper lag adjustments abnormal frequencies, change in resistance with temperature, etc.
 - **Lack of symmetry in magnetic circuit** :-In case the magnetic circuit is not symmetrical, a driving torque is produced which makes the meter creep.

ERRORS IN SINGLE PHASE ENERGY METER

- The errors caused by the braking system are,
 - (i) changes in strength of brake magnet
 - (ii) changes in disc resistance
 - (iii) self -braking effect of series magnet flux
 - (iv) abnormal friction of moving parts .

INSTRUMENT TRANSFORMER

- Instrument transformers are employed in conjunction with standard low range instruments.
- There function is to step down current or voltage to a value small enough to be measured with meters of moderate size and cost .
- Current transformers(CT) –used for measurement of current.
- Potential transformer(PT)-used for the measurement of voltage.

RATIOS OF INSTRUMENT TRANSFORMERS

- **Transformation ratio**:-It is the ratio of the magnitude of primary phasor to the secondary phasor .

$$\text{Transformation Ratio } R = \frac{| \text{primary phasor} |}{| \text{secondary phasor} |}$$
$$= \frac{\text{primary winding current}}{\text{secondary winding current}} \quad \text{for a C.T}$$

$$= \frac{\text{primary winding voltage}}{\text{secondary winding voltage}} \quad \text{for a P.T}$$

RATIOS OF INSTRUMENT TRANSFORMERS

- **Nominal Ratio**:- It is the ratio of rated primary current (voltage) to the rated secondary winding current (voltage).

$$\text{Nominal ratio } K_n = \frac{\text{rated primary winding current}}{\text{rated secondary winding current}} \quad \text{for a C.T}$$

$$= \frac{\text{rated primary winding voltage}}{\text{rated secondary winding voltage}} \quad \text{for a P.T}$$

RATIOS OF INSTRUMENT TRANSFORMERS

➤ Turns ratio

$$\text{Turns ratio } n = \frac{\text{number of turns of secondary winding}}{\text{number of turns of primary winding}} \quad \text{for a C.T}$$
$$= \frac{\text{number of turns of primary winding}}{\text{number of turns of secondary winding}} \quad \text{for a P.T}$$

➤ Ratio correction factor:- The ratio of transformation ratio to nominal ratio.

BURDEN OF AN INSTRUMENT TRANSFORMER

- The rated burden is the maximum volt-ampere loading which is permissible without errors exceeding the limits for the particular class of accuracy.

$$\text{Total secondary winding burden} = \frac{V^2}{Z} = I^2 Z$$

V=secondary winding induced voltage

I=secondary winding current

Z=impedance of secondary winding circuit including secondary winding

CURRENT TRANSFORMER

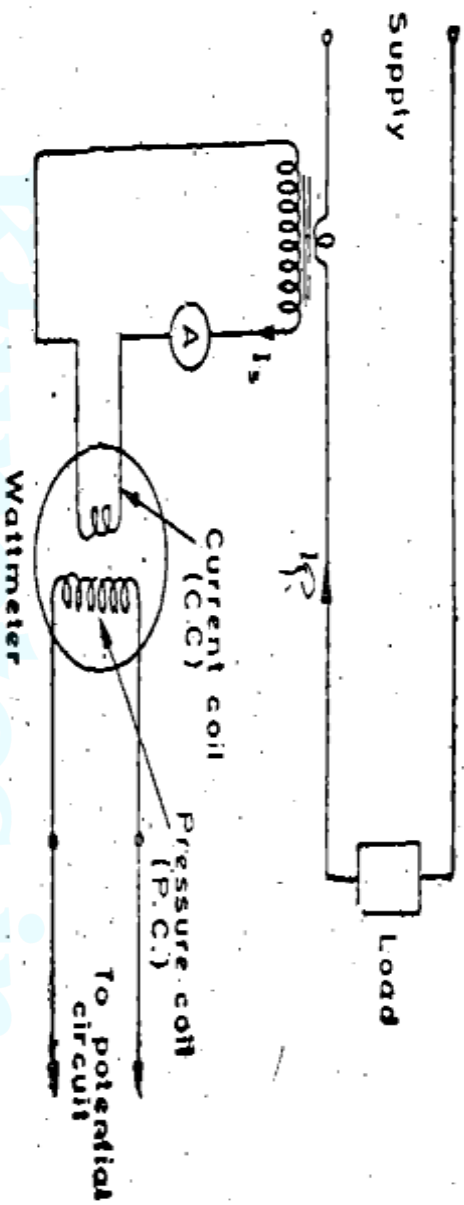


Fig. 9-3. Use of C.T. for current and power measurement .

- Used to measure current.
- Primary winding is connected in series with the line carrying the current to be measured.
- Basically it is a step up transformer.
- The primary winding consists of very few turns, and therefore there is no appreciable voltage drop across it.

CURRENT TRANSFORMER

- The secondary winding of the current transformer has larger number of turns ,its exact number is determined by the turns ratio.
- The ammeter, or wattmeter current coil are connected directly across the secondary winding terminals.
- Secondary side should never be kept open , because since it is a step up transformer secondary voltage will be high which may be dangerous to life and insulation if primary is energized.
- One of the terminals of secondary winding is earthed so as to protect equipment and personnel in the vicinity in the event of insulation breakdown.

CURRENT TRANSFORMER

10.5.1. Relationships in a Current Transformer. Fig. 10.4 represents the equivalent circuit and Fig. 10.5 the phasor diagram of a current transformer. The diagrams are same as for any other transformer.

$$n = \text{turns ratio} = \frac{\text{number of secondary winding turns}}{\text{number of primary winding turns}}$$

r_s = resistance of the secondary winding,

x_s = reactance of the secondary winding,

r_e = resistance of external burden i.e., resistance of meters, current coils etc. including leads,

x_e = reactance of external burden i.e., reactance of meters, current coils etc. including leads,

E_p = primary winding induced voltage,

E_s = secondary winding induced voltage;

N_p = number of primary winding turns,

N_s = number of secondary winding turns,

V_s = voltage at the secondary winding terminals, I_s = secondary winding current,

I_p = primary winding current,

θ = phase angle of transformer,

Φ = working flux of the transformer,

δ = angle between secondary winding induced voltage and secondary winding current,

$$\text{= phase angle of total burden including impedance of secondary winding} = \tan^{-1} \left(\frac{x_s + x_e}{r_s + r_e} \right)$$

λ = phase angle of secondary winding load circuit i.e., of external burden = $\tan^{-1} \frac{x_e}{r_e}$

I_o = exciting current,

I_m = magnetising component of exciting current, I_l = loss component of exciting current,

α = angle between exciting current I_o and working flux Φ .

CURRENT TRANSFORMER

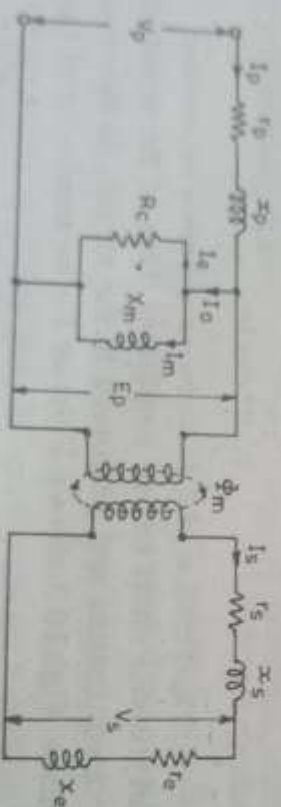


Fig. 10.4. Equivalent circuit of a C.T.

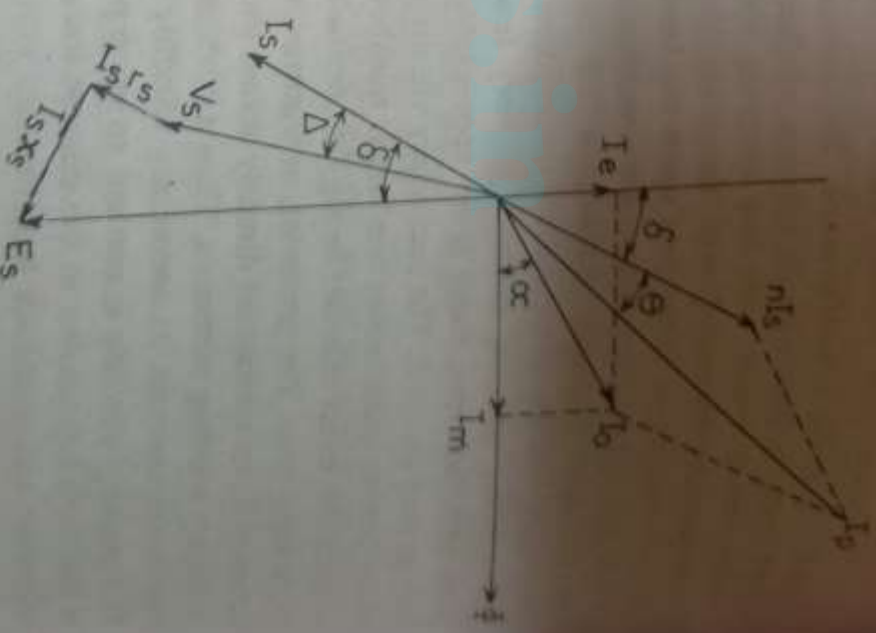


Fig. 10.5. Phasor diagram of a C.T.

CURRENT TRANSFORMER

Phase Angle. The angle by which the secondary current phasor, when reversed, differs in phase from the primary current, is known as the *phase angle* of the transformer.

This angle is taken to be + ve if the secondary current reversed leads the primary current. The angle is taken as - ve if secondary current reversed lags behind the primary current.

The angle between I_1 reversed and I_2 is θ . Therefore, the phase angle is θ .

From the phasor diagram (Fig. 10.6)

$$\tan \theta = \frac{bc}{ob} = \frac{bc}{oa + ab} = \frac{I_0 \cos (\delta + \alpha)}{nI_s + I_0 \sin (\delta + \alpha)}$$

$$\text{As } \theta \text{ is very small, we can write } \theta = \frac{I_0 \cos (\delta + \alpha)}{nI_s + I_0 \sin (\delta + \alpha)} \text{ rad.} \quad \dots (10.15)$$

Now I_0 is very small as compared to nI_s and, therefore, we can neglect the term $I_0 \sin (\delta + \alpha)$

$$\therefore \theta = \frac{I_0 \cos (\delta + \alpha)}{nI_s} \text{ rad.} \quad \dots (10.16)$$

$$\approx \frac{I_0 \cos \delta \cos \alpha - I_0 \sin \delta \sin \alpha}{nI_s} = \frac{I_m \cos \delta - I_e \sin \delta}{nI_s} \text{ rad.} \quad \dots (10.17)$$

$$\approx \frac{180}{\pi} \left(\frac{I_m \cos \delta - I_e \sin \delta}{nI_s} \right) \text{ degree.} \quad \dots (10.18)$$

ERRORS IN CURRENT TRANSFORMER

i) Ratio error:-

- From the expression for transformation ratio it can be seen that it is different from nominal ratio .
- It depends on the magnetizing current I_m , the working component I_e and the secondary current I_s .
- This difference will result in error when using a CT for measurement.

$$\% \text{ Ratio error} = \frac{k_n - R}{R} \times 100$$

ERRORS IN CURRENT TRANSFORMER

ii) Phase angle error

- Ideally secondary current must be 180° out of phase with primary current.
- From the phasor diagram, it can be seen that there is a phase difference between I_p and nI_s .

$$\theta = \frac{I_0 \cos(\delta + \alpha)}{nI_s} \text{ rad.}$$
$$\approx \frac{180}{\pi} \left(\frac{I_m \cos \delta - I_e \sin \delta}{nI_s} \right) \text{ degree.}$$

PROBLEMS

1

A CT has a single turn 1° and 200 turn 2° . The secondary winding supplies a current of 5A to a load of 1Ω resistance. The required flux is set up in the core by an mmf of 80 AT. If the frequency is 50 Hz & cross-sectional area is 100mm^2 , calculate,

- 1) Transformation ratio
- 2) Phase angle of transformer

Also find magnetic field density. Neglect iron losses, magnetic leakage & IR losses

$$\text{Transformation ratio } R = \frac{I_P}{I_S}$$

$$I_S = 5\text{ A}$$

$$\vec{I}_P = n\vec{I}_S + \vec{I}_0$$

$$I_P = \sqrt{(nI_S)^2 + (I_0)^2}$$

$$n = \frac{200}{1} = 200$$

PROBLEMS

$$\vec{I}_0 = \vec{I}_W + \vec{I}_m$$

$$I_W = 0$$

$$I_0 = I_m = \frac{\text{mmf}}{N_p} = \frac{80}{1} = 80 \text{ A}$$

$$I_P = \sqrt{(200 \times 5)^2 + 80^2} = 1003.2 \text{ A}$$

$$R = \frac{I_P}{I_s} = 200.6$$

$$\tan \theta = \left(\frac{I_m \cos \delta - I_W \sin \delta}{I_s} \right)$$

$\delta = 0 \Rightarrow$ since load is resistive

$$\tan \theta = \frac{I_m}{I_s} = \frac{80}{1000}$$

$$\theta = \tan^{-1} \left(\frac{80}{1000} \right)$$

$$= 4.574^\circ$$

PROBLEMS

$$B = \Phi / A$$

$$E_s = I_s \cdot Z_s = 5 \times 1 = 5 \text{ V}$$

$$E_s = 4.44 \Phi f N_s$$

$$\Phi = \frac{5}{4.44 \times 50 \times 200} = 1.126 \times 10^{-4} \text{ Wb}$$

$$B = \Phi / A = \frac{1.126 \times 10^{-4}}{(1000 \times 10^{-3})^2} = 0.1126 \text{ T}$$

PROBLEMS

2

A current transformer with a bar primary has 300 turns in its secondary winding. The resistance and reactance of the secondary circuit are $1.5\ \Omega$ and $1\ \Omega$ respectively including the transformer winding. With 5 A flowing in the secondary winding, the magnetizing mmf is 100 AT and the iron loss is 12 W. Determine the ratio and phase angle errors.

Solution

$$N_p = 1$$

$$N_s = 300$$

$$\text{turns ratio } n = \frac{N_s}{N_p} = 300$$

$$\text{Secondary circuit burden impedance} = \sqrt{1.5^2 + 1^2} \\ = 1.8\ \Omega$$

PROBLEMS

For secondary winding circuit,

$$\cos \delta = \frac{R}{Z} = \frac{1.5}{1.8} = 0.833 \quad \phi$$

$$\sin \delta = \frac{X}{Z} = \frac{1.0}{1.8} = 0.555$$

Secondary induced voltage $E_s = 5 \times 1.8 = 9 \text{ V}$

Primary induced voltage $E_p \Rightarrow E_s / \frac{N_s}{N_p} = \frac{9}{\frac{300}{3000}} = 90 \text{ V}$

$$E_p = \frac{9}{300} = 0.03 \text{ V}$$

$$I_e = \frac{I_{\text{iron loss}}}{E_p} = \frac{1.2}{0.03} = 40 \text{ A}$$

$$I_m = \frac{\text{magnetizing mmf}}{N_{\text{winding turns}}} = \frac{100}{1} = 100 \text{ A}$$

PROBLEMS

Actual transformation ratio $R = \frac{N I_s + I_o \sin(\delta + \alpha)}{I_s}$

$$= \frac{N + I_o \sin \delta \cos \alpha + \cos \delta \sin \alpha}{I_s}$$

$$= \frac{N + I_o \cos \delta}{I_s}$$

$$= \frac{N + I_o \cos \delta + I_m \sin \delta}{I_s}$$

$$= \frac{300 + 40 \times 0.833 + 100 \times 0.555}{5}$$

$$= 317.6$$

PROBLEMS

In the absence of any information to the contrary, we can take nominal ratio to be equal to turns ratio.

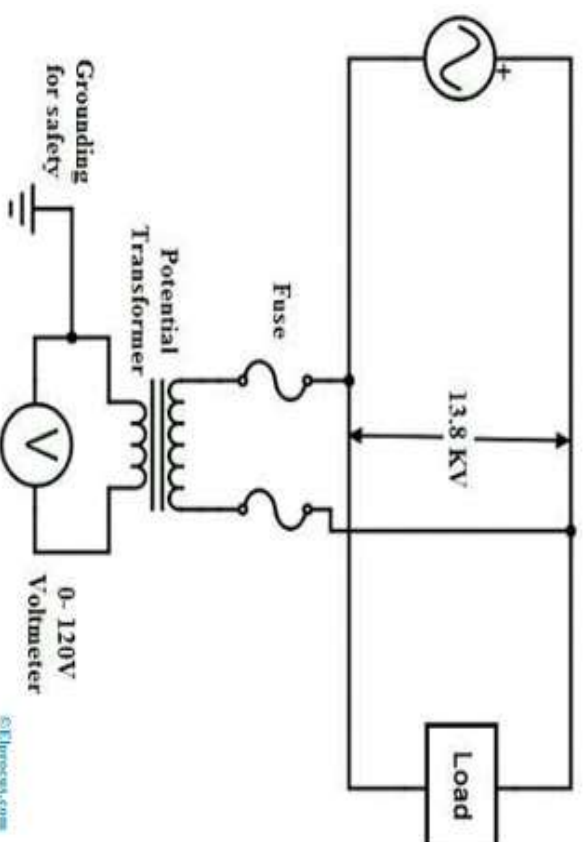
$$K_n = 300$$

$$\% \text{ Ratio error} = \frac{K_n - R}{R} \times 100 = \frac{300 - 317.6}{317.6} \times 100$$
$$= -5.54 \%$$

$$\text{Phase angle error } \theta = \frac{180}{\pi} \left(\frac{I_m \cos \delta - I_e \sin \delta}{n I_s} \right)$$
$$= \frac{180}{\pi} \left(\frac{100 \times 0.833 - 40 \times 0.555}{300 \times 5} \right)$$
$$= 2.34^\circ$$

POTENTIAL TRANSFORMER

- An instrument transformer used for the transformation of voltage from a higher value to the lower value.
- This transformer step down the voltage to a safe limit value which can be easily measured by the ordinary low voltage instrument like a voltmeter, wattmeter and watt-hour meters, etc.



POTENTIAL TRANSFORMER

- Primary winding having larger number of turns is connected across the line carrying the voltage to be measured.
- Secondary winding made of lesser number of turns is connected across the voltage circuit.

POTENTIAL TRANSFORMER

10.6.2. Relationships in a Potential Transformer. The theory of a potential transformer is essentially the same as that of a power transformer. The main point of difference is that the power loading of a P.T. is very small and consequently the exciting current is of the same order as the secondary winding current while in a power transformer the exciting current is a very small fraction of secondary winding load current.

Φ = working flux.

I_m = magnetizing component of no load (exciting) current,

I_e = iron loss component of no load (exciting) current,

I_0 = no load (exciting) current,

E_s = secondary winding induced voltage,

V_s = secondary winding terminal voltage,

N_p = primary winding turns

N_s = secondary winding turns,

I_s = secondary winding current,

r_s = resistance of secondary winding,

x_s = reactance of secondary winding,

r_e = resistance of secondary load circuit,

x_e = reactance of secondary load circuit

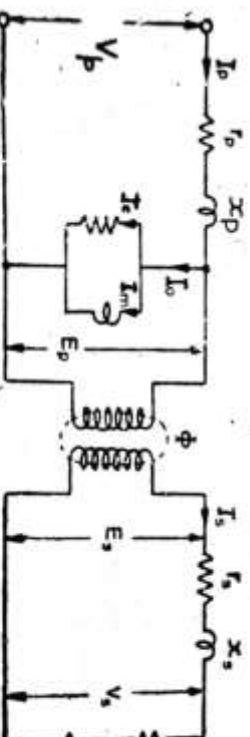


Fig. 10.20. Equivalent circuit of a potential transformer.

Δ = phase angle of secondary load circuit

E_p = primary winding induced voltage,

r_p = resistance of primary winding.

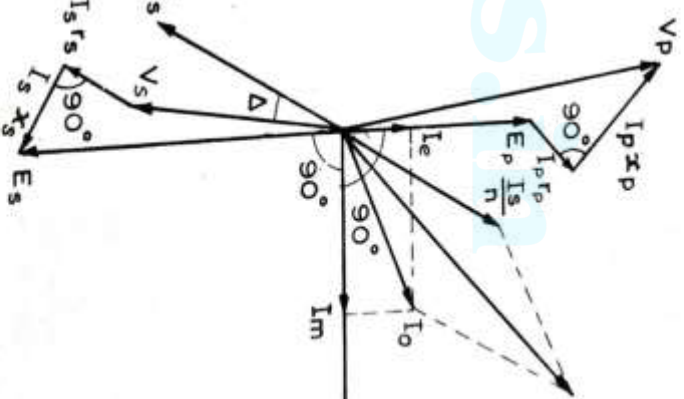


Fig. 10.21. Phasor diagram of a potential transformer

$\Delta = \tan^{-1} x_e / r_e$

I_p = primary winding current,

x_p = reactance of primary winding

POTENTIAL TRANSFORMER

R_p = equivalent resistance of the transformer referred to the primary side $= n^2 r_s + r_p$

X_p = equivalent reactance of the transformer referred to the primary side $= n^2 X_s + X_p$

$\therefore$ Actual transformation (voltage) ratio $R = V_p/V_s$

$$= n + \frac{I_s}{n} \left[R_p \cos \Delta + X_p \sin \Delta \right] + I_s r_p + I_m X_p \quad \dots (10.26)$$

POTENTIAL TRANSFORMER

Phase Angle $\tan \theta = \frac{ab}{oa} = \frac{I_p \cdot x_p \cos \beta - I_p r_p \sin \beta + n I_s x_s \cos \Delta - n I_s r_s \sin \Delta}{oa}$
 $\frac{I_s}{n} (X_p \cos \Delta - R_p \sin \Delta) + I_e x_p - I_m r_p$
 The terms in the denominator involving I_p and I_s are small and, therefore, they can be neglected as compared with $n V_s$.

$$\begin{aligned} \therefore \tan \theta &= \frac{I_p x_p \cos \beta - I_p r_p \sin \beta + n I_s x_s \cos \Delta - n I_s r_s \sin \Delta}{\frac{I_s}{n} (X_p \cos \Delta - R_p \sin \Delta) + I_e x_p - I_m r_p} \\ &= \frac{I_p \left(I_e + \frac{I_s}{n} \cos \Delta \right) - r_p \left(I_m + \frac{I_s}{n} \sin \Delta \right) + n I_s x_s \cos \Delta - n I_s r_s \sin \Delta}{\frac{I_s}{n} \left(X_p \cos \Delta - R_p \sin \Delta \right) + I_e x_p - I_m r_p} \\ &= \frac{I_s \cos \Delta \left(\frac{x_p}{n} + n x_s \right) - I_s \sin \Delta \left(\frac{r_p}{n} + n r_s \right) + I_e x_p - I_m r_s}{\frac{I_s \cos \Delta}{n} \left(x_p + n^2 x_s \right) - \frac{I_s \sin \Delta}{n} \left(r_p + n^2 r_s \right) + I_e x_p - I_m r_p} \\ &= \frac{n V_s}{\frac{I_s \cos \Delta}{n} X_p - \frac{I_s \sin \Delta}{n} R_p + I_e x_p - I_m r_p} \left[\frac{I_s}{n} (X_p \cos \Delta - R_p \sin \Delta) + I_e x_p - I_m r_p \right] \\ &= \frac{n V_s}{n V_s} \end{aligned}$$

Since θ is small, $\theta = \tan \theta$

$$\begin{aligned} \therefore \theta &= \frac{\frac{I_s}{n} (X_p \cos \Delta - R_p \sin \Delta) + I_e x_p - I_m r_p}{n V_s} \text{ rad.} \\ &= \frac{I_s}{V_s} (X_s \cos \Delta - R_s \sin \Delta) + \frac{I_e x_p - I_m r_p}{n V_s} \text{ rad.} \end{aligned} \quad \dots (10.31)$$

ERRORS IN POTENTIAL TRANSFORMER

i) Ratio error

- The actual ratio of transformation varies with operating conditions and the error may be defined as,

$$\% \text{ Ratio error} = \frac{K_n - R}{R} \times 100$$

ERRORS IN POTENTIAL TRANSFORMER

ii) Phase angle error

- Ideally secondary winding voltage must be 180° out of phase with primary winding voltage.
- From the phasor diagram, it can be seen that there is a phase difference between V_p and V_s reversed.

$$\theta = \frac{\frac{I_s}{n} (X_p \cos \Delta - R_p \sin \Delta) + I_e X_p - I_m r_p}{n V_s} \text{ rad.}$$
$$= \frac{I_s}{V_s} (X_s \cos \Delta - R_s \sin \Delta) + \frac{I_e X_p - I_m r_p}{n V_s} \text{ rad.}$$

EXTENSION OF RANGE USING INSTRUMENT TRANSFORMERS

➤ Extension of range of wattmeter using instrument transformers

- For very high voltage and current circuits, high rating wattmeters are not available . So instrument transformer can be used.

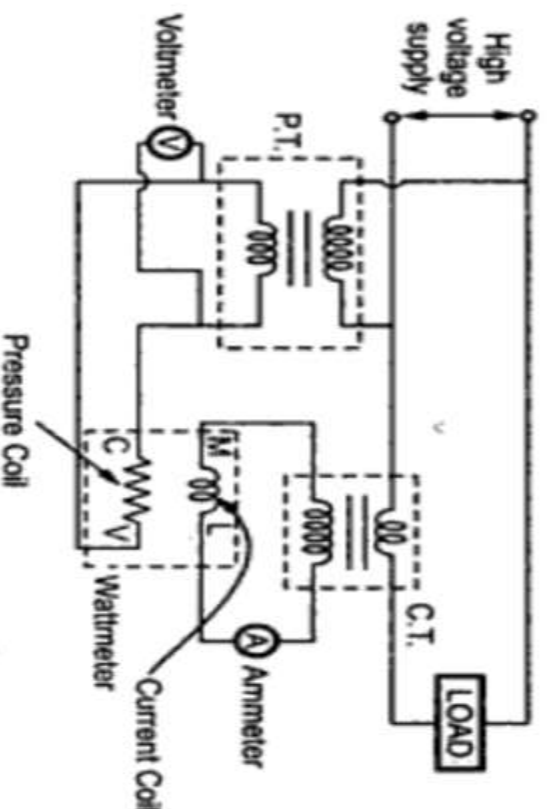


Fig. 3.35 Power measurement using C.T. and P.T.

EXTENSION OF RANGE USING INSTRUMENT TRANSFORMERS

- The primary winding of CT is connected in series with the load and secondary in series with an ammeter and current coil of wattmeter.
- The primary winding of PT is connected across the supply and secondary is connected across voltmeter and pressure coil of wattmeter.
- One secondary side of both instrument transformers are grounded.

EXTENSION OF RANGE USING INSTRUMENT TRANSFORMERS

➤ Extension of ammeter range using instrument transformer

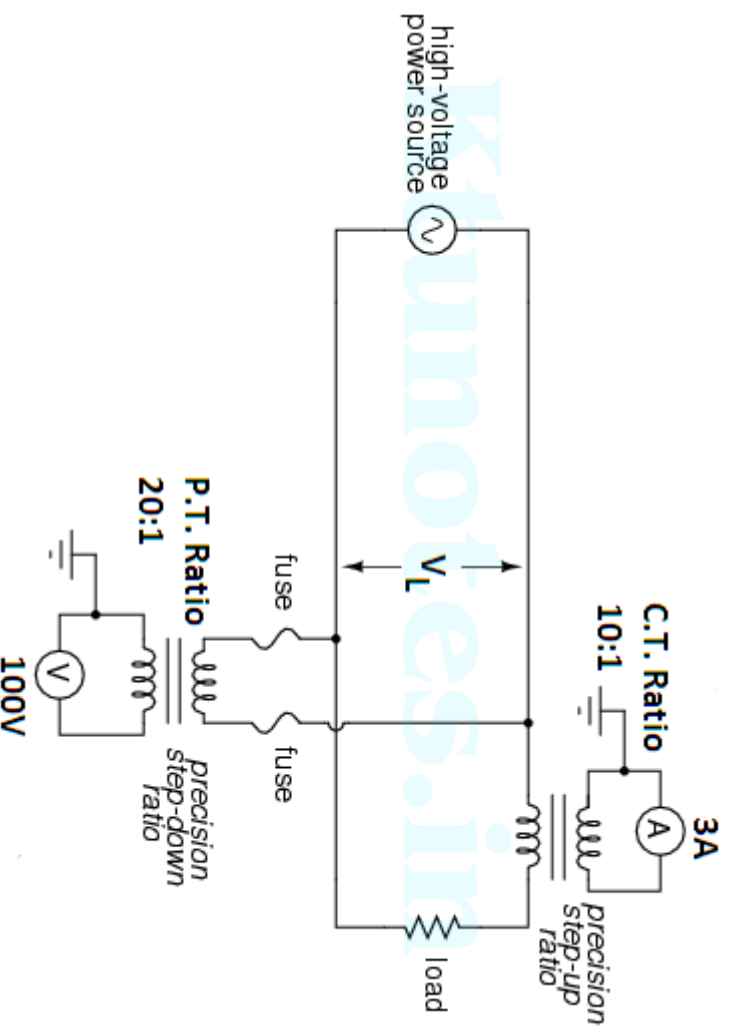
- For ranges above 0 – 250 A, a current transformer is used in conjunction with 0 – 5 A moving iron AC ammeter as shown in the figure.
- The primary of this transformer is connected in series with the load and carries the load current. The AC ammeter is connected across the secondary of the transformer.
- Since in figure, the current transformer ratio is 10:1, it means that line (or load) current is equal to 10 times the reading on the AC meter. Therefore, load current, $I_L = 3 \times 10 = 30 \text{ A}$.

EXTENSION OF RANGE USING INSTRUMENT TRANSFORMERS

➤ Range Extension of Voltmeter by Potential Transformer

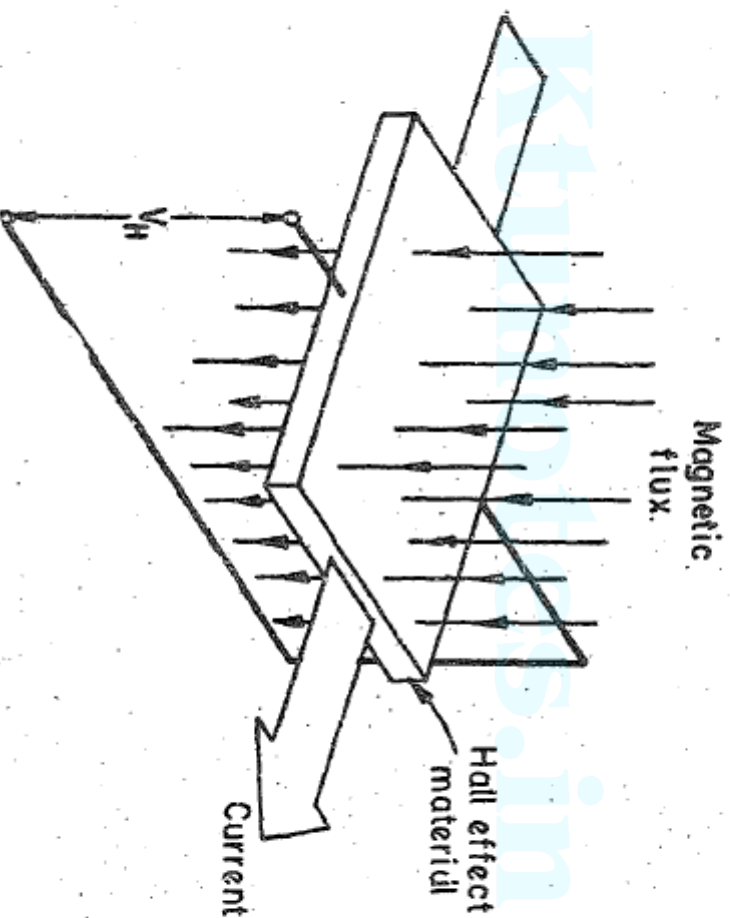
- The range of a moving-iron AC voltmeter is extended by connecting a high resistance (multiplier) in series with it.
- For ranges higher than 0 – 750 V, where power wasted in the multiplier would be excessive, a 0 – 110 V moving-iron AC voltmeter is used in conjunction with a potential transformer as shown in the figure.
- The primary of the transformer is connected across the load across which voltage is to be measured. The AC voltmeter is connected across the secondary.
- Since in figure, the potential transformer ratio is 20:1, the load voltage is equal to 20 times the reading on the AC voltmeter. Load voltage, $V_L = 100 \times 20 = 2000 \text{ V}$
- Note that both secondaries of the instrument transformers are grounded as a safety measure.

EXTENSION OF RANGE USING INSTRUMENT TRANSFORMERS



HALL EFFECT

5. **Hall Effect.** If a strip of conducting material carries current in the presence of a transverse magnetic field as shown in Fig. 6'5, an emf is produced between two edges of conductor. The magnitude of the voltage depends upon the current, flux density and a property of conductor called "Hall Effect Co-efficient"



Hall effect element

HALL EFFECT MULTIPLIER

11.7. Hall Effect Multiplier. In applications where the power is to be controlled or processed further, Hall Effect Multipliers are used. A Hall effect multiplier is shown in Fig. 11.18. The Hall effect multiplier uses a Hall effect element

The current is passed through the current coil which produces a magnetic field proportional to the current i . This field is perpendicular to the Hall effect element. A current i_p , proportional to the voltage, is passed through the Hall effect element in a direction perpendicular to field as shown. The current is limited

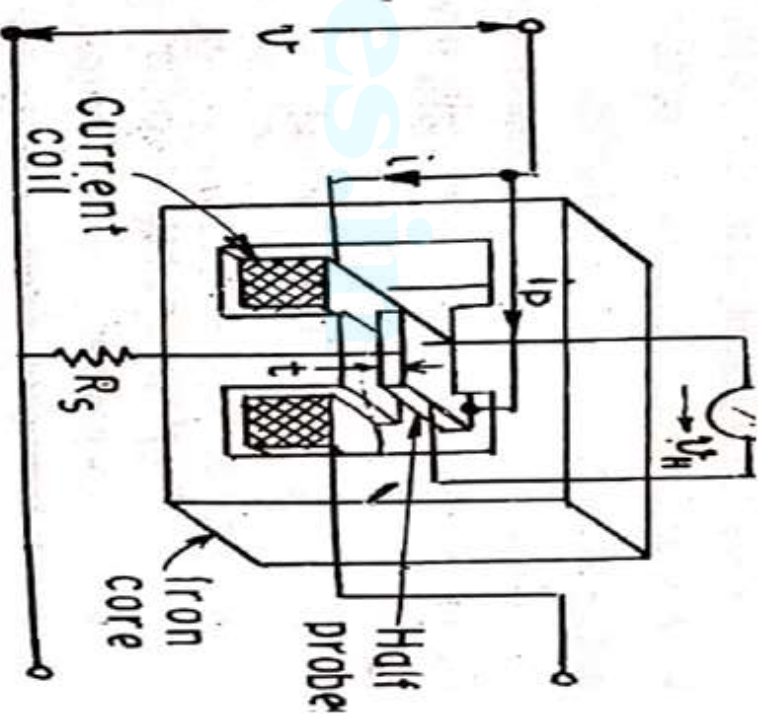


Fig. 11.18. Hall effect multiplier.

HALL EFFECT MULTIPLIER

by the multiplier resistance R_s . The output voltage of the Hall effect multiplier is :

...(11.29)

$$V_H = K_H i_p B/t$$

where K_H = Hall co-efficient ; V-m/A-Wb m⁻² ,

B = flux density ; Wb/m² , and t = thickness of Hall element ; m.

Now $B \propto i$ and $i_p = (V/R_s) \propto V$.

Hence $V_H \propto V_i$

$\therefore$ The output voltage of the Hall effect multiplier is proportional to instantaneous power.

Hence the voltmeter connected at the output terminals can be calibrated in terms of power. The Hall effect voltage which is representative of the power, can be processed further for control and other purposes. This is the major advantage of Hall effect multiplier over electrodynamicmeter wattmeters the output of the later being the deflection of a pointer which cannot be processed further.

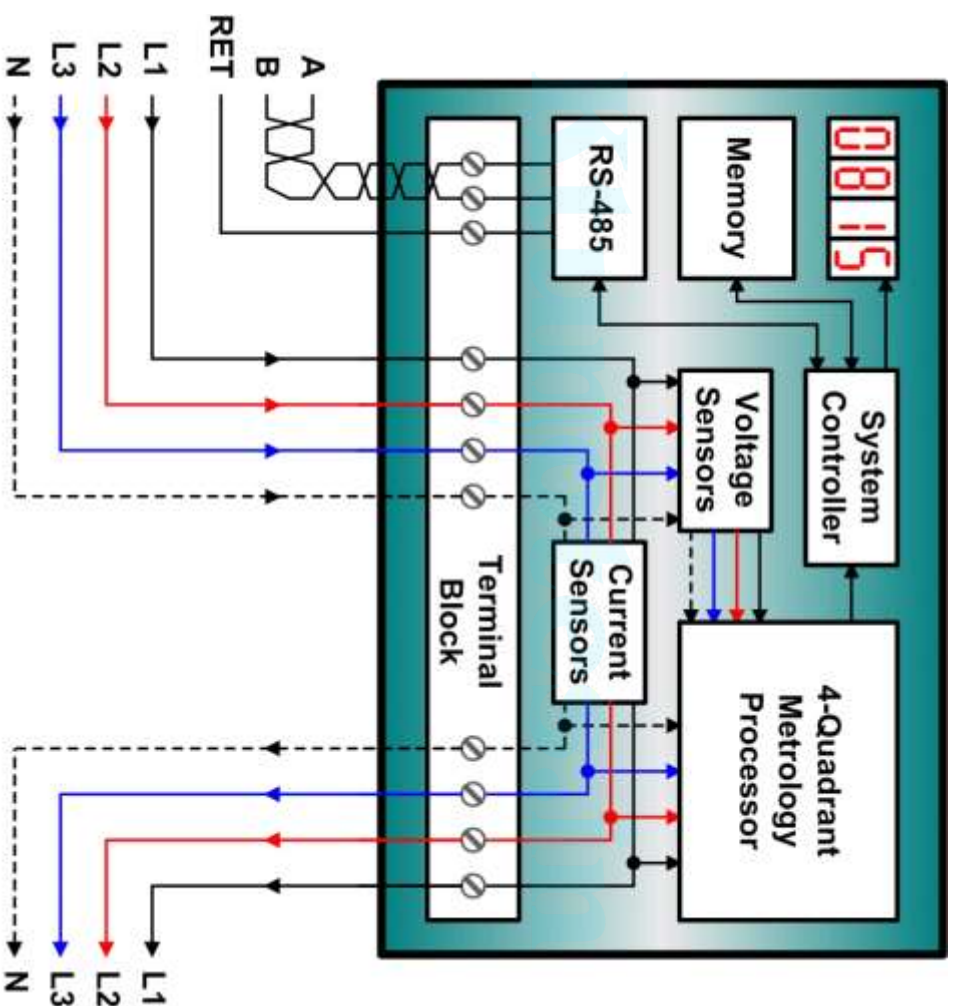
ELECTRONIC ENERGY METER

- Electronic Energy Meter is based on Digital technology and uses no moving parts. So the electronic energy meter is known as Static Energy meter.
- In Electronic energy meter the accurate functioning is controlled by a specially designed IC called Application Specified Integrated Circuit (ASIC). ASIC is constructed only for specific applications.
- The basic Electronic Energy meter senses the current and voltage signals from the circuitry, converts them into the digital signal and makes necessary calculations to get the units of electrical energy which has been consumed.

ELECTRONIC ENERGY METER

- In addition to ASIC, analog circuits, voltage transformer, current transformer etc. are also present in electronic energy meter to sample current and voltage.
- The output is converted into digital data by the analog-digital converter presented in the ASIC.
- The output of analog-digital converter is available in pulses indicated by the light emitting diode (LED) placed on the front panel of electronic energy meter. These pulses are equal to average Kilo Watt Hour (kwh\unit).

ELECTRONIC ENERGY METER



ELECTRONIC ENERGY METER

➤ Advantage of Electronic Energy Meter

- Better accuracy.
- Digital display.
- **Ease of Measurement:** With use of modern digital signal processors, it is possible to make complex calculations in a simpler way.

➤ Disadvantage of Electronic Energy Meter

- Incorrect magnitudes of fluxes.
- Incorrect phase angles.

SMART ENERGY METERS

- It is an advanced metering technology involving intelligent methods to read, process and feedback the data to customers.
- It measures energy consumption, remotely switches the supply to customers and remotely controls the maximum electricity consumption. Smart metering system uses the advanced metering infrastructure system technology for better performance.
- These are capable of communicating in both directions. They can transmit the data to the utilities like energy consumption, parameter values, alarms, etc. and also can receive information from utilities such as automatic meter reading system, reconnected/disconnected instructions, upgrading of meter software's and other important message.

SMART ENERGY METERS

- These meters reduce the need to visit while taking or reading monthly bills. Modems are used in these smart meters to facilitate communication systems such as wireless, fiber cable, power line communications.
- Another advantage of smart metering is complete avoidance of tampering of energy meter where there is scope of using power in an illegal way. Tampering means doing any act which causes the meter to run slowly or not at all and is basically theft of electricity.
- Smart meter is an electronic device that records information such as consumption of electric energy, voltage levels, current, and power factor. Smart meters communicate the information to the consumer for greater clarity of consumption behavior, and electricity suppliers for system monitoring and customer billing. Smart meters typically record energy near real-time, and report regularly, short intervals throughout the day

SMART ENERGY METERS

➤ **Advantages of Smart Energy Meters**

- Eliminating manual meter readings.
- Monitoring the electrical system more quickly.
- Providing real time data useful for balancing electrical load and reducing power outages.

➤ **Disadvantages of Smart Energy Meters**

- Difficult to manage and store vast quantities of metering data.
- Difficulty in Ensuring the security of metering data.

TOD METER

- Time Of Day (TOD) meter records the demand, time, and energy usage of electricity.
- TOD metering normally splits rates into an arrangement of multiple segments including on-peak, off-peak, and critical peak.
- The TOD metering benefits customers by providing reduced usage rates during off-peak times, which gives customers a chance to reduce their utility bill.
- TOD metering involves dividing the day into tariff slots with higher rates at peak load periods and lower tariff rates at off-peak load periods. This can be used to automatically control usage on the part of consumer.
- It is often consumers responsibility to control his own usage, or pay accordingly.